Scaling up skills-based grading for introductory syntax*

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Abstract

In an attempt to better support student learning and well-being, educators have been increasingly turning to "alternative" methods of grading. One particular approach, skills-based grading, breaks down the learning outcomes into discrete "skills" that students can demonstrate directly, within a transparent grading structure. Previous success stories for skills-based grading in linguistics required a significant amount of manual labor, and were confined to small-to-medium-sized specialist classes. We report here on an implementation of skills-based grading in the syntax component of an introductory course, where the size of the class (150-180 students) and resourcing constraints required us to scale up from existing models. Our implementation included automatically-graded online exercises, short written assignments that can be graded quickly, and an optional short assignment for deeper syntactic thinking, all of which served to make syntax more approachable. We discuss a range of quantitative and qualitative findings indicating that this approach benefits students and faculty, and conclude with recommendations for colleagues who might also be interested in centering learning and feedback in their assessments.

1 Introduction

In many institutions, introductory syntax forms a core part of undergraduate linguistic training. Though positioned as necessary in order to engage with literature across a range of sub-disciplines, syntax is also seen by many students as unnecessarily difficult or obtuse (Bailey et al. 2025). This perception is often compounded by traditional assessment structures, which provide limited opportunities for feedback or development. Combined, these factors tend to lead to high levels of anxiety and a focus on "just getting the grade". This paper reports on a project meant to address these issues through implementation of an alternative assessment structure, SKILLS-BASED GRADING (SBG), in a large UK undergraduate linguistics course. Our findings include higher student satisfaction and a greater focus on content over grades, without drastic changes to lecturer or tutor workloads.[1] Furthermore, we aim to demonstrate that ALTERNATIVE ASSESSMENT can be implemented even within the self-imposed constraints of UK "quality assurance" regimes, and that such alternatives are scalable beyond small cohorts.

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1. TUTOR is used throughout for Graduate Teaching Assistant. In keeping with UK terminology, STAFF refers to academic faculty and the support staff who work closely with them (e.g. departmental administrators). We distinguish where relevant a numerical MARK from an intuitive or letter GRADE, following Clark and Talbert (2023).

The contemporary study of syntax is typical of theoretical linguistics in that it balances scientific rigor and precise formalisms, on the one hand, with appeal to the linguist's own knowledge of their language, on the other hand. Yet in institution after institution, syntax has earned a reputation for being a particularly difficult subject. There are many possible reasons for this. One is that the analytical jump required to conceive of the sentences we utter every day as scientific objects governed by abstract rules might prove difficult. In other words, while syntax is not inherently harder than other fields of linguistics, the skills which need cultivation might sit on a sharp learning curve. A related reason is that this approach to language might not come easily to students who enter linguistics with an interest in other subfields, like sociophonetics or discourse analysis. Experience also shows that minoritized students often feel like studying syntax is a gatekeeping exercise, meant to keep them outside of "proper" linguistics (Bjorkman et al. 2023; Dockum and Green 2024). The combination of these factors has led to syntax often being thought of as "the math of linguistics": a formal subject which students are often told implicitly or explicitly that they aren't good at (Coles and Sinclair 2022; Bailey et al. 2025).

For the reasons given, we wanted to re-evaluate the way we teach an introductory syntax block. However, changes are unlikely to be implemented or sustained if they result in higher initial costs or heavier workloads for staff involved. This is especially true in the UK higher education context, where—at the time of writing—universities are implementing historic funding cuts that divert resources away from the development of teaching and learning (McCann 2024; Senatus Academicus 2025; McAlpine 2026). A key concern of this project is therefore the identification of alternative grading principles and practices that can address issues with negligible or no increases in resource demands when replicated at scale. To this end, the area identified for easiest change within undergraduate syntax training was assessment and grading, with a focus on scale and automation where possible.

We report on the combined experience of two iterations of the course, in the academic years 2024/25 and 2025/26, drawing primarily on the most recent run. We begin in Section 2 with some background on assessment and SBG. Section 3 describes our course, and Section 4 provides quantitative and qualitative findings. Section 5 concludes.

2 Alternative grading, evidence-based assessment, and SBG

In various models of learning, making an error, identifying the error, and taking steps to resolve the resulting incongruity form an integral part of the learning process (Piaget 1952; VanLehn 1988; National Research Council 2000; Ambrose et al. 2010). Given that mistakes are a fundamental (and inevitable) part of any learning process, they should be sought out, not penalized; the learning benefits of reflecting on identified errors with feedback are supported by the experimental literature (VanLehn et al. 2003; Butterfield and Metcalfe 2006; Potts and Shanks 2014; Cyr and Anderson 2015; Metcalfe 2017). However, under common approaches to assessment and grading, when a student's coursework shows us where they can improve, we punish them by docking marks (Blum 2020); and worse, we deprive them of the opportunity to demonstrate that they have improved, making the punishment permanent.

Traditional assessment typically requires students to submit summative one-off coursework, which is assigned a numerical mark. The assessment may be split across multiple instances, such as mid-term and final exams, but what characterizes these forms of assessment is their summative nature. Typically, the material taught before a mid-term assessment is not systematically revisited or retested, and students have little opportunity to demonstrate subsequent learning or growth in response to feedback.[2] Furthermore, how such marks are arrived at is not always transparent (see, for example, discussion in O'Leary and Stockwell (2022) of the issues around partial credit in semantics). The disadvantages of such an approach have

2. Resit assessments and remedies are often employed, but as described in Zuraw et al. (2019:e406), these workarounds are "patches on a fundamentally flawed system".

been highlighted repeatedly, especially when contrasted with alternatives emphasising learning, feedback and growth (Butler 1988; Kohn 1993/2018; Lipnevich and Smith 2008; Schinske and Tanner 2014; Sackstein 2015; Blum 2020; Gannon 2020; Clark and Talbert 2023; Kastner 2023; Stommel 2023; Eyler 2024).

In response to this incongruity, many educators are increasingly turning to "alternative" or "authentic" forms of assessment instead, including SBG. Some of these alternatives include pass/fail, project-based learning (Sambell et al. 2013), self-assessment or "ungrading" (Blum 2020; Gorichanaz 2022; Sorensen-Unruh 2024), mastery learning (Bloom 1968), specifications grading (Nilson 2014), and standards- or skills-based grading (Buckmiller et al. 2017; Clark and Talbert 2023; Iannone et al. 2025).[3] Common themes among these approaches include reducing student anxiety through low-stakes or no-stakes assessment, constructive alignment between learning outcomes and assessment (Biggs and Tang 2011), centering the role of making errors and receiving feedback (Ambrose et al. 2010; Sambell et al. 2013), and promoting equity or the reduction of bias in grading (Feldman 2019). A growing pedagogical movement follows the four principles outlined by Clark and Talbert (2023), who advocate for clearly defined standards, helpful feedback, marks as indicating progress on a transparent scale, and multiple (but not unlimited) reattempts without penalty. We discuss these aspects of SBG next, followed by existing implementations within linguistics.

2.1 Skills-based grading

In SBG, instructors formulate in advance fine-grained learning outcomes as SKILLS which students are expected to demonstrate throughout the course. Students submit work that demonstrates mastery of the skills, which are then graded as *proficient* or *not yet proficient* for each student. Other variants are also possible, such as adding *almost proficient*, which counts as *not yet proficient* and is only for feedback purposes (Zuraw et al. 2019; O'Leary and Stockwell 2021, 2022; Clark and Talbert 2023), and *advanced proficiency*, which is used for calculating final "A" grades as set apart from "B" and lower grades (Zuraw et al. 2019). Students have multiple attempts at any given skill, allowing them the time to learn from previous feedback. There is no penalty for repeated attempts at a skill, and once demonstrated a skill cannot be "lost" in any meaningful sense. The counterpoint to this is that there is no requirement to attempt any given exercise unless the student needs to demonstrate that skill.[4]

This approach incorporates two factors key to addressing the issues with syntax teaching and assessment outlined above. First, it increases marking transparency. Grading individual skills is binary, making it easier and clear. Calculating the final mark then relies on a rubric where marks correlate with the number or complexity of skills (ours is described in Section 3.2). Second, it allows multiple "low stakes" attempts at any skill. Materials and exercises teach and probe these skills directly. Students are given multiple chances to demonstrate competence in a given skill, through homework exercises and tests, with no penalty for repeat attempts. This kind of approach emphasizes learning over the attainment of grades, as failure is de-stigmatized and students are given clear evidence of their own progress when they resolve a previous error and demonstrate a skill previously marked *not yet*. It also means that graders can assign *not yet* to a student submission, safe in the knowledge that students can get feedback and try again. SBG has been argued to result in lower levels of stress and a renewed focus on the material rather than the grade, for staff as well as students (Buckmiller et al. 2017; Lewis 2022). Additionally, although SBG cannot remove bias

3. Skills-based grading and standards-based grading are used here to refer to the same general approach. For consistency with other implementations in linguistics (Zuraw et al. 2019; O'Leary and Stockwell 2021, 2022), we use skills-based grading. See Clark and Talbert (2023) for further nuance.

4. In some SBG implementations, it is necessary for students to demonstrate skills multiple times. One example is given in O'Leary and Stockwell (2022), where the same skill must be demonstrated four times to be marked as proficient. This results from a coarser grained wording in the skill, such that it is necessary to apply the skill in multiple contexts to truly demonstrate proficiency. It is also possible to write in more granular terms for closely related skills each requiring individual demonstration; see Pankratz (2026) for discussion of the relevant levels of granularity.

in assessment, core features have been shown to reduce certain types of bias and structural disadvantages (Quinn 2020; Reibel 2022; Yik et al. 2025).

2.2 SBG in linguistics

Within linguistics, SBG has been used successfully in phonetics, phonology (Zuraw et al. 2019), semantics (O'Leary and Stockwell 2022), and medium-sized syntax courses (Brian Hsu, p.c.). The skills used in these courses cover a range of technical and abstract reasoning abilities, for example transcribing segments (Zuraw et al. 2019:e426) and calculating semantic type (O'Leary and Stockwell 2021:871). Many of the skills practiced in the earlier parts of a course are also entailed by later skills. For example, using islands as a diagnostic for movement assumes an understanding of constituency structure.

In explaining the choice to adopt a skills-based framework, Zuraw et al. (2019) give the list of requirements in (1). The first two (1a,b) align clearly with the issues of student anxiety and broader perceptions about syntax that were on our minds, and the logistical constraints placed on assessment reforms identified above. The third speaks to the intended learning outcomes.

(1) Requirements of a skills-based system:

a. Grades should reflect whether students eventually mastered the material, not how long it took them to master it.

b. Grading should be easy. Each question would be marked either right or wrong, with no partial credit.

c. There should be a meaningful distinction between simple mastery—a B grade—and going beyond that level—an A grade. (Zuraw et al. 2019:407)

The requirement in (1a) is addressed in a skills-based framework by increased transparency. Students can attain a skill at any point throughout the course and are encouraged to make mistakes and learn from them. Two students may demonstrate the same skill at very different points in the course, depending on when each student was ready. Many of us are familiar with the example of a student for whom the basics of a formalism only "click" at office hours towards the end of the semester, when their mid-term exam had already demotivated them and they are scrambling to prepare for the final. This is a situation we would want to avoid.

The requirement in (1b) is key in present context. As mentioned above, the current UK higher education context means that any assessment reform that requires time is frowned upon, even if a short-term increase in costs pays off in the long term. SBG simplifies grading to a binary distinction. Not only does this remove the uncertainty from partial credit scores, it also suggests that at least some parts of the assessment in SBG can be automated. However, in the linguistics courses noted above, instructors and tutors marked all of the assessment themselves by hand. A legitimate criticism is that such labor-intensive practices are unfeasible on courses with large cohorts and high student-to-staff ratios. We were interested in piloting SBG in a setting where completely manual marking would be prohibitively costly.

Zuraw et al. (2019) identify two logistical issues specifically relevant to attempts to scale SBG upwards. First, the design and implementation of the gradebook can directly impact the workload for staff. Although they report grading being faster overall, they describe the use of an "elaborate" Excel spreadsheet on their courses as time consuming (Zuraw et al. 2019:e419). On courses with more students and more staff, direct entry of marks into a shared Excel spreadsheet by hand will undoubtedly be more burdensome. Second, a key feature of SBG is that students can track their own progress. Marks for a given skill need to be communicated to students in some way (such as an online gradebook or spreadsheet), and this communication needs to be fast enough that students are able to devote time to learning and re-attempt the skill.

Requirement (1c) raises the question of what it means to "think like a linguist" beyond learning a specific formalism. In a typical introductory course in theoretical linguistics, we want to train students in a common formal framework while also teaching them the conceptual skills that transcend any given formalism. The basic skills of syntactic argumentation, for example, are relevant regardless of whether one happens to be working in Minimalism, Lexical-Functional Grammar or Combinatory Categorial Grammar.

A skills-based approach allows instructors to make this distinction explicit. To satisfy the requirement in (1c), the phonology course of Zuraw et al. (2019) defined a number of difficult questions which allowed students to demonstrate understanding beyond the requirements of the course, that is, to show that they can think like phonologists. Additionally, answers to other questions that demonstrated a deeper understanding than what was required were also marked *Advanced Proficiency*. Accumulation of *Advanced Proficiency* marks allowed students to attain A grades for the course.

In the phonetics course in Zuraw et al. (2019) and the semantics course in O'Leary and Stockwell (2022), a different approach was taken. Separate skills were defined, *Apply* and *Scientific Analysis* respectively, and demonstration of these skills was required for letter A grades but not lower letter grades. These can be understood in terms of the levels of Bloom's taxonomy (Anderson and Krathwohl 2001). We discuss our own implementation of SBG in what follows, including our separation of narrower formal skills from broader skills, the latter also leading to an "A" grade.

3 LEL2A syntax

The current project aimed to implement SBG within the introductory syntax block of a second-year undergraduate course on the linguistics of English at a UK university. While students in previous iterations had continued on to more advanced courses, doing well in them, they consistently noted that the syntax block of this module was particularly challenging. Similarly, when students disengaged—as happens on any course—this often happened in the syntax block. We give the relevant background on the course first, before turning to assessment design and implementation.

3.1 The course

LEL2A: Linguistic Theory and the Structure of English is a 2nd-year undergraduate course taught at the University of Edinburgh. LEL2A is a required course for linguistics and related degree programs, but also has a non-negligible proportion of visiting students and students from other degree programs each year (roughly 10–15%). It is a semester 1 course that students take in their second year of four.

The course follows on from more general introduction to linguists courses in year 1 by looking in more depth at the linguistics of English. Students are introduced to formal approaches to phonology, morphology, syntax and semantics. The course is a pre-requisite for more specialized courses in years 3 and 4, but also for the year 2, semester 2, course *LEL2D: Cross-linguistic Variation, Limits and Theories*, where students are introduced to advanced formal analysis of cross-linguistic variation.

Initial enrollment each year is between 150-180 students. The course runs for 11 weeks, with three lectures per week and one small-group tutorial. Prior to implementation of the SBG framework, the course was divided between the four blocks as in (2).

(2) The four blocks of LEL2A, in order:
   a. Phonology, 3 weeks
   b. Morphology, 1 week
   c. Syntax, 5 weeks (reduced to 4 weeks with the implementation of SBG)
   d. Semantics, 2 weeks (increased to 3 weeks with the implementation of SBG)

Assessment for the phonology and morphology blocks was combined, but syntax and semantics had separate summative assessments. The three assessments are independent of each other and combine to give a weighted average for the course.

As noted, tutor and student feedback traditionally identified a lack of engagement in the syntax block, coupled with an increase in student anxiety. Our goal was to reorient the assessment for learning, and to do so without increasing resource demands. We did not set out to modify in any substantial way the content of the syntax block, which focuses on the basics of X-Bar Theory, although we did reduce the time devoted to the syntax block from five weeks to four after re-organizing our materials. This also allowed a more balanced treatment of the other blocks.

The topics covered in the new syntax block are given in (3). These topics were split over 12 lectures (three per week over 4 weeks). This translated into 11 TOPICS plus one mid-block consolidation session.[5] Students were directed to the Santorini and Kroch (2007) textbook for further readings.

(3) Topics in the syntax block
- a. Topic 1 – What is Syntax?
- b. Topic 2 – Constituent Structure and Constituency Tests
- c. Topic 3 – Predicates and arguments: syntactic and semantic arguments
- d. Topic 4 – Predicates and arguments: arguments and modifiers, subjects
- e. Topic 5 – X-bar and the clause: X-bar schema
- f. Topic 6 – X-bar and the clause: IP, modals, V and I, VPISH, CP
- g. Topic 7 – Nonverbal XPs: Arguments of N & the DP Hypothesis
- h. Topic 8 – Passives
- i. Topic 9 – Non-finite clauses Part 1: To-infinitives, Raising and Control
- j. Topic 10 – Non-finite clauses Part 2: Why raise & Object Control
- k. Topic 11 – *Wh*-questions (bonus)

3.2 Assessment structure

The SBG assessment structure for the course was created from a review of the existing course materials for the previous few years. We conceptualized the intended learning outcomes as formal SYNTAX-SKILLS and broader RESEARCH-SKILLS, described below, following the general principles of backward design (McTighe and Wiggins 2004). Attainment of these was mapped on to a rubric for the final mark, given in Table 1. Marks are aligned to the UK system, in which 40 is a pass and 70 is an “A”.

As mentioned, the content of the course was left largely unchanged from previous years. This meant that a wealth of materials already existed. Additionally, a dedicated Master’s course in our department, *Introduction to Syntax*, covers much of the same content. Both courses had existing lecture notes, quizzes, and past papers. All that was required was aligning these to the topics and skills in a process of backward design. For this, two PhD students, previous tutors on the course (one syntactician and one discourse analyst), were hired as research assistants to go through the existing materials and tag them with potential skills. A pairing of one syntactician and one non-syntactician was specifically chosen to help ensure that the questions and instructions were transparent and accessible. The tagged material was then edited into notes and question banks aligned with each skill.

5. Between the first and second SBG iterations of the course, some topics were re-ordered to introduce passives before raising and control. Additionally, in the first iteration of the course, there were originally twelve topics. Non-verbal XPs were split across two topics with more dedicated skills. This was reduced mid-course to add the consolidation session in response to student feedback.
6. Showing mastery of at least one skill from each of Topics 2–6, or an equivalent for each Topic from later skills.

LEL2A syntax mark	Syntax Skills	Research Skills
85	16/18	5/5
80	16/18	4/5
75	16/18	3/5
70	14/18	3/5
65	14/18	—
60	12/18	—
55	10/18	—
50	8/18	—
40 (pass)	6/18[6]	—
30 (fail)	3/18	—

Table 1: Grading rubric for the syntax block

The two research assistants were contracted for 35 hours in total, divided between the two of them, for a total of just over 700 GBP.[7] Additional funds were used on book vouchers for volunteer focus group students to test a preliminary version of the course materials, and on literature.

After identifying what had been taught and what had been assessed in previous iterations of the course, a list of syntax-specific skills was identified, such as those in (4). These were initially edited down to 24 discrete skills, spread over 12 topics. Following feedback from the first cohort and student focus groups, this was further cut down to a total of 18 syntax skills over 11 topics.[8] These are, in effect, learning outcomes for each topic. The full list is available in Appendix A.

(4) Examples of syntax-skills:

a. Skill 2a: Perform an appropriate constituency test for an NP.

b. Skill 4b: Perform an appropriate diagnostic to distinguish a complement from an adjunct.

c. Skill 5b: Use X' representations to distinguish adjunction from complementation.

As noted above, assessment was meant to provide feedback, allow multiple attempts, provide transparent reflection of attainment, and be low-stakes. Assessing syntax-skills therefore explicitly tagged which skill or skills were being targeted in a given question. These skills could be demonstrated in two ways: first, in discrete automatically-graded exercises on our online learning management system, each targeting just one skill (up to two attempts allowed). And second, in four integrated short-form tutorial submissions graded by the tutors, in which questions combined a number of skills (also explicitly tagged). Each skill was graded as Proficient or Not Yet Proficient, with feedback provided at the tutorial and through answer keys. In total, then, students had three signposted opportunities to demonstrate mastery: two attempts at discrete online exercises (no deadline & instant feedback) and one integrated tutorial sheet per week (deadline & delayed live feedback). Once demonstrated, a skill could not be "un-demonstrated" or lost, either in exercises for the same topic or in subsequent weeks.

To illustrate, consider the skills for Topics 5 and 6 in Figure 1, corresponding to the 2nd and 3rd lectures in the second week of the block. These four skills cover different aspects of syntactic analysis within the X-bar schema. Skill 5a establishes the basics of the schema: heads, phrases, specifiers, and how they all

7. Our budget came from discretionary funds available to our department. There are no dedicated university funds for improving individual courses. For comparison, an external consultancy firm contracted by our university to improve "efficiency" was paid over 200,000 GBP per year (FOI 2025/00680), and the external consultancy firm that carried out a survey for a "Student Vision" was paid around 70,000 GBP (all-staff town hall, November 2024). A recent contract acquiring access to two additional sets of Large Language Models is worth over 500,000 GBP (national reference APR554901).

8. This paring down of skills is typical in the process of distilling the most appropriate set of skills (Clark and Talbert 2023). The bonus Topic included a 19th skill which students could demonstrate and which would count towards their total.

fit together. Skill 5b implements the distinction between arguments and adjuncts (established in Skill 4b independently of the formalism). Skill 6a extends the system with additional functional heads, and Skill 6b ties it all together by moving the VP-internal subject from Skill 5a to the specifier of IP from Skill 6a.[9] We show how the system was implemented in what follows, using the example of Skill 5b.

Topic 5 – X-bar and the clause: X-bar schema

5a Represent heads and phrases, identified via constituency tests, in a manner consistent with the *projection principle.*

5b Use X′ representations to distinguish adjunction from complementation.

Topic 6 – X-bar and the clause: IP, modals, V and I, VPISH, CP

6a Use constituency tests to identify and justify projection of different functional heads in a representation, such as I, C, and NEG.

6b Implement a *VP-internal subject hypothesis* analysis.

Figure 1: The syntax-skills for Topics 5 and 6.

3.3 Implementing syntax-skills

The Learning Management System in our institution is Anthology's Blackboard Learn Ultra, and outside of this we have limited access to other systems. This created a series of constraints as it was important to keep the set-up as simple and scalable as possible while meeting the SBG requirements. As there was no scope in the project to obtain additional software or resources, the priorities for learning technology were to minimize complexity for staff and students, use the existing systems, and prioritize sustainability and re-use.[10] We go into detail here in order to showcase the pedagogical aspects, but also to explain how a similar design can be implemented in this or other learning management systems.

3.3.1 Individual exercises

Each syntax-skill had a corresponding "Learn Test" (exercise) in Blackboard, drawn from the question banks created by the two research assistants. Each exercise contained a pool of between 3–5 automatically graded questions, one of which was randomly selected when an exercise was launched. The exercise was configured to allow for two attempts to be taken by each student, encouraging the learner to undertake the exercise again, should they not have managed to demonstrate proficiency first time round; if a first attempt did not demonstrate proficiency, the student could take it again with a different question.[11] The due date was set as the final date of the semester, so that students could undertake the assessment at their own pace.

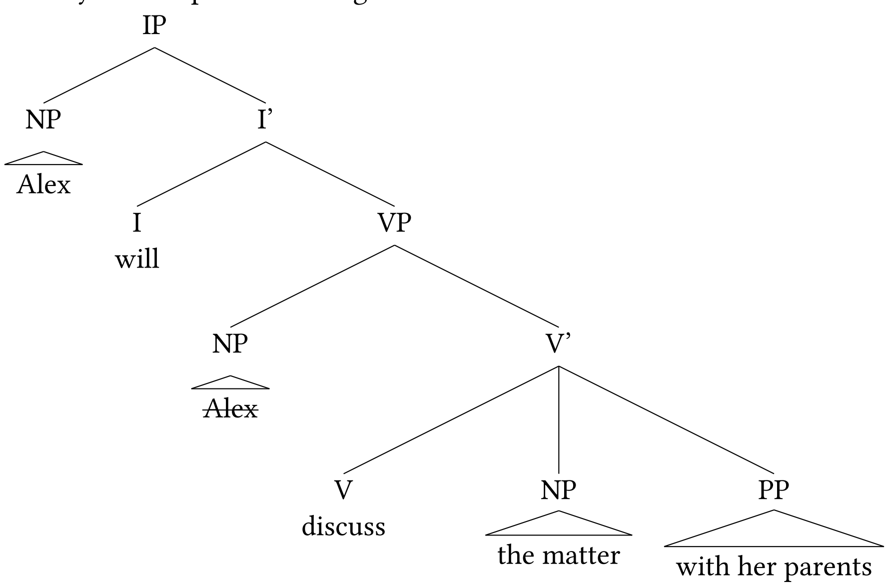
9. We use Inflection Phrase (IP) rather than the perhaps more common Tense Phrase (TP) for historical reasons.

10. Some of the non-VLE options considered were to make use of SharePoint, using lists to track student skills acquired, as well as the University-run Digital Badge scheme. None of these options fully responded to the objectives and, as a result, the decision was made to keep the process entirely within Blackboard.

11. While this was our original intention, Blackboard could not ensure that the second question picked from the pool was different from the first one.

An example of one of the questions for Skill 5b is given in Figure 2. Students are presented with a tree that does not distinguish between the argumenthood of the two phrases following the verb. The student must use the constituency tests developed earlier, in Skill 4b, to test the claims that can be read off of the tree.[12] After submitting their response, students received automated feedback, prepared in advance, which discussed the questions in more detail. As the exercises are automatically graded, tutor or instructor input was not required.

[Skill 5b]
The following tree fails to account for one or more acceptability judgments which would typically be held by native speakers of English.



Which of the judgements below are **inconsistent with the tree** (not predicted by it) as it is given?

(5) a. Alex will discuss the matter with her parents, but I will not.

b. Alex will discuss the matter with her parents, and I will do so with my parents.

c. Who will Alex discuss the matter with?

d. *Alex will discuss the matter with her parents, and I will do so the matter with my siblings.

Figure 2: A Blackboard exercise for syntax-skill 5b.

The exercises were linked to a specifically created grading schema imported into the online Gradebook with the grades Proficient and Not Yet Proficient. Upon completing an exercise, the student could immediately see whether they had obtained the skill, which was then registered in the Gradebook. Each exercise was linked via a naming convention to the different skills (in this case 5b).

3.3.2 Synthesizing tutorial sheets

The written part of the assessment allowed the students to show mastery through the completion of a tutorial sheet containing a number of longer questions. These questions cover multiple topics and skills,

12. 67% of students answered this question correctly, with the remaining third of responses split fairly evenly between the three distractors.

and could be achieved in lieu of or alongside the single-skill exercises.

Figure 3 reproduces an excerpt from the second tutorial sheet, due at the end of the second week of the block (after Topic 6). Question 3 here combines aspects of three different skills, which are all flagged explicitly. This question does a number of things. First, it gives the student another opportunity to show proficiency in these skills. And second, it introduces slight complications which would not preclude a student from obtaining the skill, but which can lead to fruitful discussion: sentence (6) contains multiple adjuncts, which students would not have drawn in class, and which can lead to discussion of recursion, iteration and scope. Sentence (7) combines negation with a ditransitive verb.

Question 3 5b ☐, 6a ☐, 6b ☐

Q3 Returning to the sentences given in (6) and (7), repeated below, draw trees to represent them. One tree is enough to demonstrate the skills, but you should try both: each has an extra challenge!

(6) John gave Bill the book quickly in the street on Tuesday.

(7) John will not put the book on the table.

Figure 3: Tutorial question for syntax-skills 5b, 6a and 6b.

Students were required to upload a PDF file with their answers, which they had unlimited attempts to submit on Blackboard. However, the tutorial sheet did have a due date, as it required tutors to review the submission: in our case, by Friday afternoon for tutorials on Monday.

Tutors worked with their own tutorial groups throughout the semester. In order to let tutors see submissions from the students in their groups, Blackboard's functionality of DELEGATED MARKING was added to the set-up. This feature requires that the tutorial groups be specified once per tutor on the system, after which a student's submission could go straight to their tutor. Tutors would only see relevant submissions, reducing the administrative load inherent in distributing assignments from 150 students.

Tutors marked the submissions for their own small group tutorials and tagged skills as Proficient or Not Yet Proficient in Blackboard, again using the imported schema. Concretely, they noted down the obtained skills on the Blackboard feedback for the tutorial submission, and also cross-checked the Gradebook; if a student did not yet have a Proficient grade for that skill, the tutor marked it as Proficient. They were asked to do so by the end of the week (within five days). A copy of these instructions is included with the Supplementary Materials.

Written feedback on the submissions was minimal, consisting primarily of obtained skills. Instead, the tutorials themselves discussed the material, and an answer sheet was made available after all tutorials had run. Student could not "lose" a Proficient grade already obtained, regardless of how they performed on the tutorial sheets.

To summarize, students had three attempts to demonstrate proficiency in the discrete syntax-skills, distributed in any way the student liked between two attempts at individual, automatically graded Blackboard exercises, and more holistic written answers to tutorial sheets.

3.4 Implementing research-skills

Given that students should be trained in how to "think like a linguist" or "think like a syntactician" independently of a specific formalism, and since we had to abide by a marking scheme that included "A" grades for noteworthy performance, we designed an assessment that allows students to explore aspects of syntax beyond those covered in the class material directly.

To do this, we included 5 general Research-Skills that students could write up for a 70+ mark ("A" grade), shown in (8). These are analogous to the *Apply* and *Scientific Analysis* skills from Zuraw et al. (2019) and O'Leary and Stockwell (2022). This 1-2 page submission was more open-ended, giving students a chance to introduce their own data, explore further literature or carry out additional research. Submission of the research skills assessment was optional, and students were encouraged to focus on the syntax-skills first. In line with the methodology adopted, two submission dates were provided, giving students the option to revise and resubmit if desired. The first draft could be submitted for feedback and grading by the end of week 10 of the semester. Just over half of the students submitted the optional coursework, which was marked by two of the six tutors.

(8) Research-skills in the syntax block
  a. Skill M1: State a hypothesis clearly
  b. Skill M2: Present data in support of a hypothesis
  c. Skill M3: Define the conditions under which a hypothesis would be refuted
  d. Skill M4: Present data to refute a hypothesis
  e. Skill M5: Consider the limitations of a diagnostic test when interpreting the results

A brief was provided that defined and exemplified the 5 research skills with examples drawn from the course content (see Supplementary Materials). The brief provided three suggested topics but also encouraged students to get in contact to suggest their own topic. The suggested topics directed students to tackle problems that were left unsolved during the course, such as how to analyze ditransitive structures or split intransitivity, or extending the material covered in class to other languages. Although students were encouraged to apply the syntax-skills they had learned in the course, the submission was intended as an opportunity to "do syntax" without worrying necessarily about the formalism; demonstration of the skills was not conditional on using any particular formal approach. Demonstration of 3 out of 5 research skills was required for a 70+ mark (see the rubric in Table 1).

4 Results and evaluation

Evaluating the success of any teaching intervention is a non-trivial task, as it inevitably requires the use of various metrics, none of which can tell the entire story on their own (Biesta 2010, 2020). We present here a combination of quantitative and qualitative findings, relating to student performance and feedback (Section 4.1) as well as tutor impressions (Section 4.2). The study received ethics approval from the ethics review board.

4.1 Student performance

4.1.1 Syntax-skills

We first consider the total number of syntax-skills achieved by students. This metric is evaluated at two time points: the end of the syntax block (mid-November), and the end of the semester (mid-December). Assessment design prioritizing learning from feedback encourages making and correcting mistakes. This means that students should have opportunities to let the material "sink in". In LEL2A, four weeks separated the end of the syntax block from the end of the semester, at which point all coursework must have been submitted. We expected students to do moderately well in terms of demonstrated skills by the end of the teaching block, and even better by the end of the semester.

This prediction was confirmed, as can be seen in the syntax-skills histogram in Figure 4. On the x axis we see the total number of skills obtained by a student; the y axis gives the number of students who obtained that number of skills. Each total of syntax skills is presented at two points in time: at the end of the teaching block (left bar of each pair, salmon) and the end of the semester (right bar of each pair, blue).

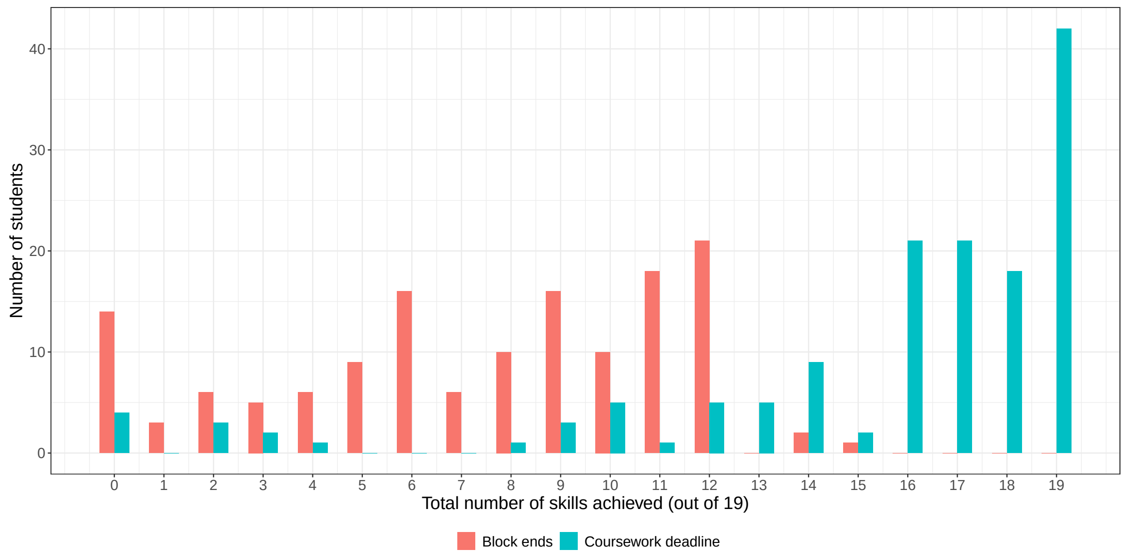


Figure 4: Number of students (y axis) who obtained the number of syntax-skills (x axis) at the end of the teaching block (salmon, left of pair) and end of semester (blue, right of pair).

A number of things are worth pointing out here. The first is that students did improve their attainment; the plot shifts from a right skew to a left skew. The second is that students were intentional about the number of skills they aimed to demonstrate. By the end of the semester, the modal value is 19, the most possible skills; in other words, many students were able to demonstrate all skills. But many students also stopped at 16 (and 17), which is exactly the number of skills required for an A grade (see 1). Students therefore had a greater degree of agency than in traditional assessment, and were not stuck in a cycle of spending large amounts of time on assignments that matter little for their final grade (not that the grade is what matters).

The same picture appears with the distribution of final marks for the block—again at these two time points—in Figure 5: not only was attainment high, but lower-performing students caught up with the material. The ceiling mark of 65 at the first time point reflects the fact that the research-skills assignment, which is necessary for a mark of 70+, was not submitted until later in the semester.

Appendix B presents a comparison of marks with previous versions of the course. While a direct comparison is uninformative as the course was taught in different ways, it does bring out two general points which we return to in the Conclusion: that it is not obvious what a numerical mark actually conveys, and that the two SBG iterations showed remarkably similar distributions.

Another way of gauging student engagement with the course is through the proportion of submitted tutorial sheets. Even though students are not required to submit these sheets, being able to obtain all of the syntax-skills through the online exercises (and vice versa), many did submit these sheets, as can be seen in Table 2. In fact, students were explicitly encouraged to submit these sheets so that they can deepen their knowledge and learn from mistakes without penalty, a challenge many took up. The limitations of Blackboard make it difficult to conduct more in-depth comparisons of exercises and tutorial sheets.

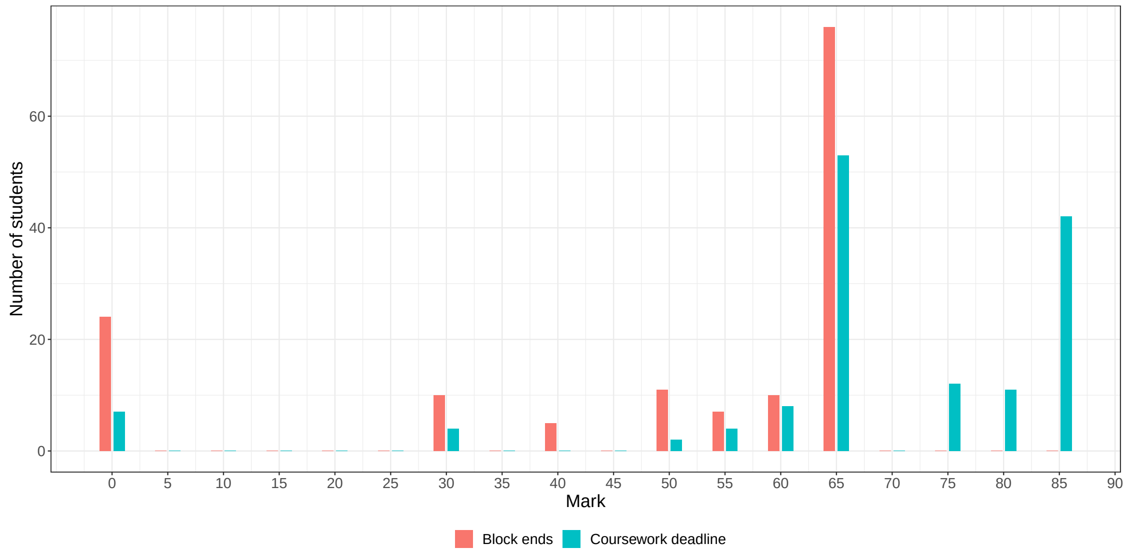


Figure 5: Number of students (y axis) who obtained the marks (x axis) at the end of the teaching block (salmon, left of pair) and end of semester (blue, right of pair).

Year	Tutorial 1	Tutorial 2	Tutorial 3	Tutorial 4
2024–25	0.76	0.57	0.48	0.37
2025–26	0.79	0.65	0.54	0.53

Table 2: Proportion of students who submitted each tutorial sheet.

4.1.2 Research-skills

Apart from the framework-specific syntax-skills, students were encouraged to develop the more general research-skills. Students could submit a first draft for feedback; 34 students did so, with almost a third of them getting 5/5. Another 37 students submitted for the final draft (or maintained the submission from the first draft).

To visualize the improvement from first to second draft, the alluvial plot in Figure 6 presents first and second submission (for those students who submitted a final attempt). The number of research-skills (out of 5) is represented on the right-hand and left-hand sides. The right-hand side presents the state of play after the final drafts were graded, and the thickness of the connecting line represents the number of students who improved from the grade on the left to the one on the right. For example, about half of the students who did not submit a first attempt ("none" on the left-hand side) nevertheless got a 5 on the second attempt ("5" on the right-hand side; 59% of all final submissions got a 5/5). Of the students who got a 2 on the first draft, roughly half improved to a 4 and half improved to a 5. Since skills could not be lost once obtained, and since the first draft counted for demonstration of research-skills, no student received a lower grade for their second attempt. What this plot demonstrates, then, is progression over time: students were able to improve their submission based on the feedback, revising and resubmitting without penalty.

4.1.3 Student feedback

Student feedback on the assessment system was positive. The official feedback from the 2025/26 student representatives in our department's biannual Student-Staff Liaison Committee read: "Students enjoy the teaching, find the submissions manageable and any concerns were resolved through consultation times

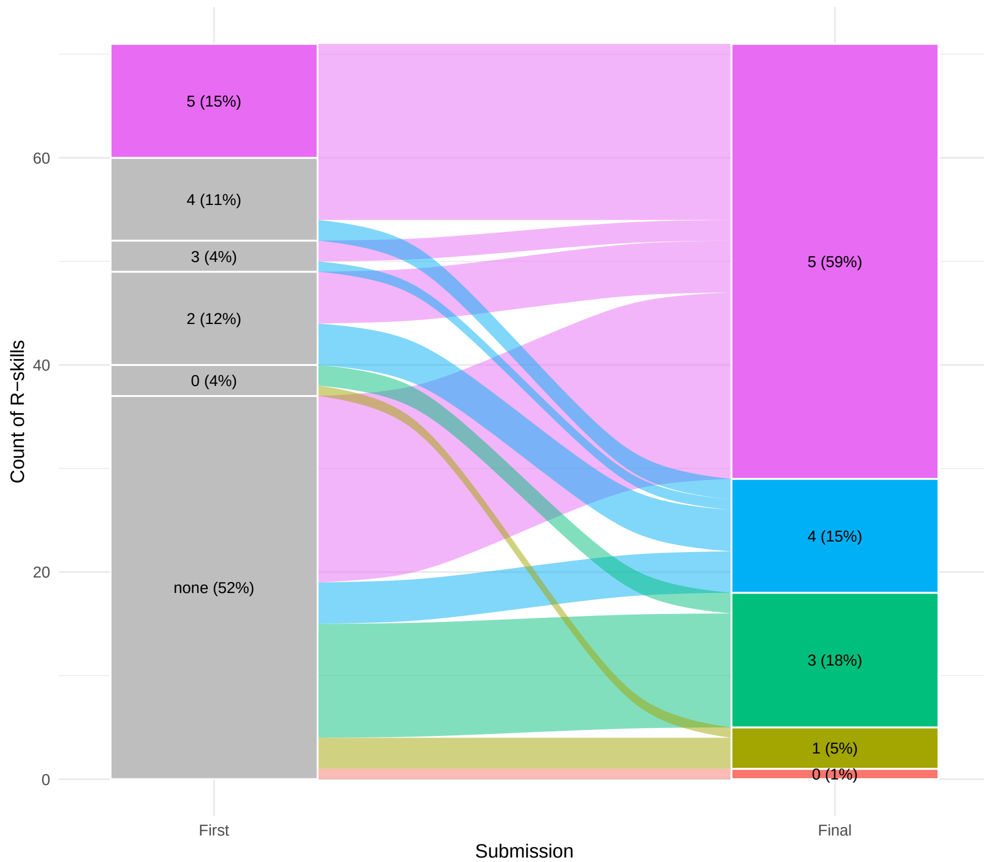


Figure 6: Research-skills obtained in the first draft and final draft. The maximum 5/5 is highlighted in the first draft for clarity.

available."

An anonymous in-class survey was carried out halfway through the block; its results are given in Figure 7, again showing high satisfaction.

A similar survey was sent out at the end of the semester. Only 18 students answered this one, and of these, only five gave free text responses (not a surprising result given the "survey fatigue" our students report, and the lack of structured institutional follow-up on student voice). These responses are given in their entirety in (9).

(9) End-of-semester student feedback

a. "I think the LEL2A syntax block had a great structure (haha see what i did there) — I never felt like I was lagging behind, and the readings (course notes + other readings) were very helpful to clarify uncertainties even before the lectures themselves. The lectures also had a good pace. Assessment structure was great as well. I felt that I finally understood syntax for the first time, and more importantly enjoyed it."

b. "The grading system was a little complicated and I still have some trouble understanding everything (for example why for the marking scheme of the skills 16 is the highest but 19 skills exist) but I do think it's a great system, taking away anxiety from students. I like that the learning effect was the focus of this block and not the grading, which made it much more enjoyable."

c. "I appreciated the different assessment style. Definitely eased nerves related to essays and re-

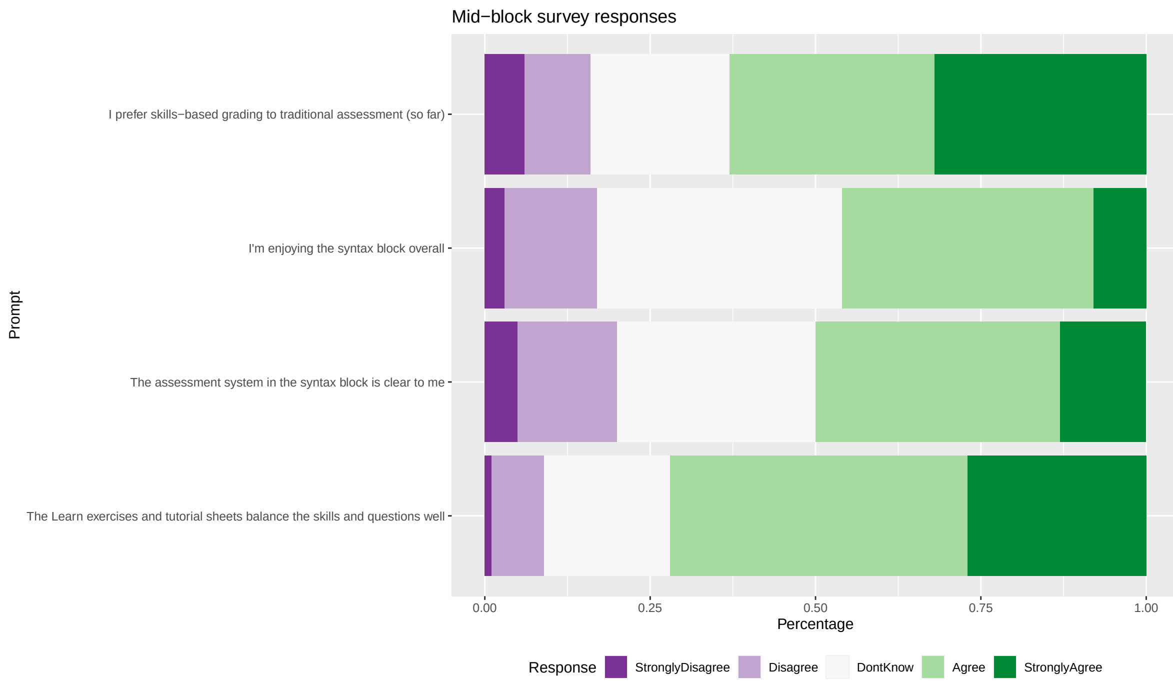


Figure 7: Mid-block in-class survey results, from "strongly disagree" to "strongly agree". N = 71.

search"

d. "The spaced repetition of completing the learn quizzes after the lectures and then completing the tutorial assignments each weekend was very helpful in reinforcing syntax content."

e. "I liked the backup of submitt[ing] all missing skills at the end of the block"

The responses in (9), while few in number, are consistent with the many informal comments we received from students in 2024/25 and 2025/26.[13] For example, students told us on multiple occasions that the course felt more collaborative. One group of friends told us that they worked on their exercises in a shared study space, and that whenever one of them achieved "Proficient" on a syntax-skill, the rest of them would cheer. Comparable insights from the tutors on the course are given in Section 4.2.

4.1.4 Follow-up courses

One more way of evaluating the syntax block is by looking at its potential effects on later courses. Metrics like these are tentative at best, but we discuss them here for completeness; they are at the very least consistent with SBG leading to better student performance and an open attitude towards syntax.

We consider one qualitative indicator in the follow-up year 2, semester 2 course *LEL2D: Cross-linguistic variation: limits and theories*. Whereas LEL2A focuses on analyzing contemporary English and on learning specific formalisms, LEL2D engages with cross-linguistic variation and the challenges it poses to our theories. This course, like LEL2A, is made up of a number of thematic blocks. One of these is morphosyntax. Qualitatively, the LEL2D teaching team reported that students were making different mistakes in the last two years (the LEL2A SBG years). Impressionistically, they were making "better" mistakes. For example,

13. The in-class component of the 2024/25 piloted a "flipped classroom" set-up, which proved too time-consuming for students and which was modified halfway through that block. Remarkably, even students who disapproved of the flipped lectures praised the assessment structure in the 2024/25 survey.

in the past, students might've analyzed a sentence in a language like Gaelic by giving a standard LEL2A-English tree with Gaelic lexical items, effectively drawing English trees for non-English examples. But now, they were more likely to attempt a different tree, and even if it was incorrect in some way, it at least showed that they were asking the right questions. Again, these impressions are somewhat diffuse, but are at least consistent with the rest of our findings. Developing effective ways to evaluate student growth throughout the curriculum would be a valuable project within linguistics.

We would have liked to obtain a quantitative measure of success by comparing the marks on the final assignment for the LEL2D morphosyntax block in the two years before SBG was implemented in LEL2A and the two years in which students had gone through SBG syntax. Unfortunately, institutional workload constraints render access to this data prohibitively difficult at this time. We would have also liked to consider in our analyses the 3rd/4th year elective *Syntax: Theory and Practice*, comparing the proportion of students who chose to enroll in the course after taking SBG LEL2A, compared to the two years prior. Here, too, generating these reports is not currently possible.

4.1.5 A note on gender attainment gaps

As discussed in Section 2.1, one reason to adopt alternative assessment practices is to create more equitable assessment. While the rationale is clear, how exactly to evaluate success is a topic of active discussion. Yik et al. (2025) evaluated the outcomes of students in a number of chemistry courses under traditional and specifications grading, concentrating on the difference in attainment between systematically advantaged and systematically disadvantaged students (according to race, gender, income, and so on). We conducted a partial replication of their study; while our institution gathers similar data, it is not curated or made available in a way that would allow similar analyses. Instead, we ran a post-hoc analysis using the most easily available characteristic, namely gender – with a null result.

In our institution, female undergraduates outnumber male undergraduates. A higher proportion of female students achieves a high degree (1st or 2:1 in the UK system) than male students do (`https://equality-diversity.ed.ac.uk/sites/default/files/2026-05/EDI%20Student%20Data%20Report%202026.pdf`). However, given general biases in society, and the stigmas associated with syntax in particular, we wanted to check whether there was any difference in our course as well.

We considered marks in the last four iterations of the course (two traditional and two SBG). We coded individual students based on student records and our own familiarity with them primarily as male or female. Non-binary students and students for whom we had no records (for example visiting students who had left the institution) were excluded from this basic analysis, removing 3.4% of the data. There was no significant difference in the overall mark for the course between female and male students, as measured by a Wilcoxon rank-sum test. There was also no significant difference in the marks for the syntax block between the female and male students, regardless of assessment regime. Unsurprisingly, then, no difference was identified between gender gaps in the syntax block, on the one hand, and gender gaps in the other blocks combined, on the other hand. Our findings therefore cannot speak to any specific claim in the literature; plots are given in Appendix C.

4.1.6 Interim summary

Implementing SBG helped student performance and created a better learning environment by all measures we tried, quantitative as well as qualitative. We had initially planned to survey students on their assessment anxiety before the syntax block and as it related to the syntax block, predicting a drop in anxiety as reported by Lewis (2022). While this is the pattern we observed, only 13 students filled out the final survey, and so we relegate this finding to Appendix D.

4.2 Tutor feedback

The small-group tutorials were divided between six tutors on each of the 2024/25 and 2025/26 iterations. Tutors had between one and four groups, with the median number of groups being two. Only two tutors taught on both iterations of the SBG version of the course, but eight had experience with the course prior to the SBG update. All of the tutors had experience tutoring on other courses within the department.

After the entire course had finished, tutors not involved in the course redesign were invited to give feedback on their experience through an anonymous survey which included a number of Likert-scale statements to rate and optional free text comments.

Given the small number of responses, the numerical responses are not particularly informative. However, when considered alongside the free response comments, consistent patterns emerge. When comparing marking under the SGB approach with the more traditional assessment employed previously, the majority of tutors preferred marking under a skills-based approach. Reasons cited for the preference for SBG related to *clarity* and *certainty* when marking:

(10) a. "I felt I had to make rather arbitrary distinctions to assign [other course's] grades, which was not the case here."

b. "*Met* or *not met* is clearer than [...] *This felt like a 70*."

Relevant to the main aims of the project, most tutors found that SBG marking took either the time allotted or less time, with many sources of variation:

(11) a. "It made the workload easier as a tutor"

b. "It took a little less time [...] I went over time with the [phonology] essays and a little under time with the weekly syntax marking. Also, if I'd done 2A syntax for the first time the syntax marking would have taken me about the allotted time."

c. "The marking itself was not a problem but because [Blackboard] can take soooo long to load I spent a lot of time waiting to mark - obviously [Blackboard] taking a minute to load is much more impactful when you are given 6 minutes a script."

d. "I think it depends on how familiar you are with the system. At the beginning it took more time, but then as the course progressed it became easier, and consequently faster."

These responses identify a learning curve for tutors marking under the new assessment framework. The tutors who identified the marking as taking longer than the allotted time wrote in the comments that this was only true at the start. The comments suggest that time saved may increase with the length of the block or course. It is customary in our department to hold weekly tutorial briefings, which provide an opportunity to co-ordinate tutorials for a given week (including grading, where relevant). In the present context, the SBG assessment structure was applied only to a four week block. By the time tutors were familiar with the framework, it was almost over. Nevertheless, it is instructive to compare the current set-up with traditional grading, where a large amount of calibration and "moderation" is needed to ensure anything approaching parity of evaluation (Lipnevich et al. 2020).

The most important benefit of SBG can be seen in the improvements to in-person tutorials and the communication between tutors and students. Tutors reported that the new assessment structure had a positive impact on their tutorials:

(12) a. "I loved that I had a very very good handle of where my students were at. I usually had a look at the weekly submissions just before the tutorial [...] I could really tailor the tutorial to the students"

b. "The weekly exercises really allowed me to see which areas my various groups were struggling with so I could tailor the tutorials to the specific needs of a group."

c. "I think the main benefit of the sbg is that I know that where the students struggled in the tutorial material before the tutorial"

d. "[I]t allows for more needs-based tutoring and a better understanding of students."

On a given week, tutorial sheets were due by Friday afternoon for a tutorial on the following Monday. This means that some tutors were able to use the students' submissions when planning the tutorials, which effectively became one-hour feedback sessions guided by the demonstrated needs of the students in the room.

Relatedly, tutors described the flow of information as going both ways, in that students were engaging with feedback and using it constructively:

(13) a. "Students read feedback."

b. "I think the increased clarity on what was required was definitely a bonus for the students, and so was their ability to watch their progress as it happened and to work flexibly in a way that suited their own priorities and schedules."

One tutor also sent us additional thoughts on the research-skills assignment:

(14) "I really like this assignment - I have read some interesting and creative answers. Also, I feel like this is a great way of encouraging them to pursue syntax further. I didn't take advanced syntax in my BA because I didn't see how you could ask research questions about syntax. I think if I'd had something like this in intro/intermediate syntax I would have felt much comfortable taking syntax further."

And finally, as would be expected from "alternative" or compassionate assessment, the resulting atmosphere improved as well:

(15) a. "[T]hey understood the system and then the pressure came down a bit."

b. "I think the syntax-skills sheet made it really clear what the tutorial was about, both in the sense that students could rely on it to understand their progress, and for planning for tutorials around the specific skill."

c. "Students were more relaxed about syntax and less panicked, especially once they understood the assessment format. Especially students that had a normal to harder time but engaged with the course benefited from SBG."

5 Conclusion

5.1 Summary and remaining issues

In our implementation of skills-based grading for introductory syntax, discrete learning outcomes ("syntax-skills") could be demonstrated either in automatically-marked exercises or in a tutorial sheet. One demonstration was enough, but in total three attempts were allowed. Students were also able to show deeper understanding via an optional assignment, again aligned to discrete skills ("research-skills"). This assessment design seemed to achieve its goals of making assessment a tool in service of learning, rather than the other way around. It goes beyond what we have seen elsewhere in linguistics in being implemented successfully with large classes (150 students), while requiring relatively little manual feedback from graders.

One aspect we have not managed to automate is the calculation of the final mark. Producing the marks according to the rubric required the instructor to export the Gradebook to a spreadsheet and construct elaborate functions there, importing the final mark back into the system. This issue is also related to Blackboard's inability to align skills with exercises, which was something we had to do ourselves by setting up separate Gradebook entries for each skill. Learning management systems that are designed with alternative assessment in mind, such as TeachFront, might be better suited for pedagogical goals.

Overall, we identified very few cases of "academic misconduct", including the use of "generative artificial intelligence" (GenAI). We believe that this is because the principles of alternative and compassionate assessment neutralize the incentives that lead to "cheating" in higher education (Bertram Gallant and Rettinger 2025). Where cases did arise—in a handful of tutorial submissions—it was easy to spot trees generated by e.g. ChatGPT which looked nothing like the tree structures discussed in class: they contained other bracketing and labels. The student was not given the skills if they did not demonstrate them in the way they were expected to. But in addition, we explained to the student that they should follow the material from class, and that they will be free to pursue other directions for the research-skill assignment. Framing the issue as one of correctly adhering to our assessment design, rather than the misuse of GenAI, allowed for less adversarial conversations.

5.2 Comparison with traditional assessment

Teaching linguistics often requires a compromise between teaching material, including specific formalisms, and teaching broader skills. How individual instructors resolve this tension can lead to interesting innovations (e.g. Harrigan et al. 2022), yet more often than not it makes us prioritize the formal analyses. When teaching syntax, we want to train students in how to think like a syntactician while teaching them a certain formalism. Some students might understand the ideas while struggling with the formalism. How this situation is resolved is a question for individual instructors and departments. For us, it meant two things. First, that a student not excited about syntax could aim for, and achieve, a decent mark (50–60) without tying themselves into knots. And second, that they can flex their syntactic muscles in a research-skills assignment, where they could use any formalism or tackle any syntactic question (within reason). Identifying these two components of the block strikes us as a clear benefit compared to traditional assessment, though it is not necessarily in conflict with traditional assessment either.

Shifting the focus away from getting a high mark actually allowed us to think about grades more intentionally. In past iterations, students received a traditional problem set consisting of English sentences that needed tree structures. They drew their trees, submitted them, and got a mark on them. But what did these grades actually convey? If some marks are docked for getting something like binary branching wrong, and a different amount is docked for not drawing V-to-T movement, what is the actual numerical basis that any of this is based on? What does a passing grade mean in terms of how much syntax the student knows? How much syntax does a student know if they have a 90 compared to an 80? While the problems inherent to grades do not disappear under SBG, the rubric is more transparent: marks indicate progress, understood as the number of concepts a student can be assumed to be comfortable with. Our course also speaks to a related misconception of alternative assessment, in which grades are higher because standards are lowered. That view is inconsistent with the clear mapping out of learning outcomes which students achieve.

An additional advantage of this backward design is that we aligned what we taught with what we assessed. In previous years, lecture material and readings covered much more material than was ever examined. Naturally, we should always have more things to show students than what we require them to do. At the same time, there is a degree of anxiety associated with not knowing what material you will be examined on. For both students and instructors, then, SBG helped maintain focus.

5.3 Recommendations and conclusion

We end with some recommendations on how to get started. Between the current paper, the linguistics courses discussed in Section 2.2, and the broader scholarship of teaching and learning, linguists now have many models to fashion our courses after. We would recommend thinking about the main learning outcomes and working backward from them, thinking how they should best be assessed, and then what materials need to be compiled in order to bring students to that point.

Our focus on design and implementation throughout this article shouldn't detract from the need to communicate with students. As mentioned earlier, we ran focus groups in which former students gave feedback on the course design before it ran for the first time. And for the students taking the course, we sought to scaffold SBG rather than introduce it all at once. SBG was introduced incrementally: we first discussed SBG with the class at the end of the first lecture, when the instructor walked through an exercise for Skill 1a and solved it together with the class. Granted, this meant that some students will have seen the answer to an exercise they will encounter later on Blackboard themselves; this was intentional, as it lowered the barrier to entry (Clark and Talbert 2023) and also helped signal that we were interested in helping students learn in a transparent system, not compete against each other. We similarly showed the tutorial sheet at the end of the last lecture of each week, walking through the questions briefly (without discussing the answers). The research-skills assignment was introduced in the middle of the third week, once students were comfortable with the syntax-skills component of the block. Throughout, we were open about the novelty of SBG in our context, posted links to relevant literature on Blackboard, and encouraged students to report any issues with the exercises. We tried to solve these as quickly as possible, giving students the benefit of the doubt when it came to e.g. unclear wording on automatically graded multiple-choice questions.

Furthermore, guidance was provided that encouraged students to space attempts from each other and to make the most of live tutorial feedback and lectures. In addition to the three signposted attempts, students were given the opportunity to identify if they had demonstrated a missed skill in a later tutorial submission. For example, a student who had not yet demonstrated Skill 2a on NP constituency could in the fourth tutorial submission perform a constituency test for an NP and identify this as demonstrating the missing skill. As with the assessment of skills, all course materials were also explicitly tagged with relevant skills, allowing feedback to direct students to specific resources.

So, what does it take to scale SBG up for large classes? Converting our traditional assessment took time and money, while also being fairly straightforward and paying off fairly quickly. On the one hand, it clearly required resources and effort. On the other hand, we mostly repurposed existing materials in a way that felt more authentic to "doing syntax". One important thing that happens with implementing any form of alternative assessment, including SBG, is more deliberate consideration of what we're assessing. That is our main recommendation, before any implementational issues: to think carefully about what we want to teach, what values we want to honor in our work, and how assessment can best serve these goals.

A Skill List

Research Skills

M1. State a hypothesis clearly.

M2. Present data in support of a hypothesis.

M3. Define the conditions under which a hypothesis would be refuted.

M4. Present data to refute a hypothesis.

M5. Consider the limitations of a diagnostic test when interpreting the results.

Syntax Skills

Topic 1 – What is Syntax?

1a. Use a hierarchical structure to represent constituency.

Topic 2 – Constituent Structure and Constituency Tests

2a. Perform an appropriate constituency test for an NP.

2b. Perform an appropriate constituency test for a VP.

2c. Perform an appropriate constituency test for PP/AP/AdvP.

Topic 3 – Predicates and arguments: syntactic and semantic arguments

3a. Identify divergences in the mapping between semantic and syntactic predicates & arguments.

3b. Identify arguments that bear semantic roles (such as agent, theme/patient, goal, & location) and any arguments that do not.

Topic 4 – Predicates and arguments: arguments and modifiers, subjects

4a. Relate differences in types of verbal complements to c-selection properties of verbs.

4b. Perform an appropriate diagnostic to distinguish a verbal complement from an adjunct.

Topic 5 – X-bar and the clause: X-bar schema

5a. Represent heads and phrases, identified via constituency tests, in a manner consistent with the *projection principle*.

5b. Use X' representations to distinguish adjunction from complementation.

Topic 6 – X-bar and the clause: IP, modals, V and I, VPISH, CP

6a. Use constituency tests to identify and justify projection of different functional heads in a representation, such as I, C, and NEG.

6b. Implement a *VP-internal subject hypothesis* analysis.

Topic 7 – Nonverbal XPs Part 1: Arguments of N & the DP Hypothesis

7a. Implement an analysis of complex noun phrases that is consistent with the framework developed.

7b. Identify elements as complements or adjuncts of a noun in a principled way.

Topic 8 – Passives

8a. Account for the behaviour of passive movement, including any relevant restrictions, within the framework developed.

Topic 9 – Non-finite clauses Part 1: To-infinitives, Raising & Control

9a. Identify non-finite clauses headed by infinitival *to* in a principled way.

9b. Distinguish raising structures from control structures on the basis of θ-role assignment.

Topic 10 – Non-finite clauses Part 2: Why raise & Object Control

10a. Motivate raising to subject position (A-movement) in terms of either CASE or the EPP.

Topic 11 – Wh-questions (guest lecture, bonus skill)

11a. Implement a movement based analysis of wh-movement.

B Marks

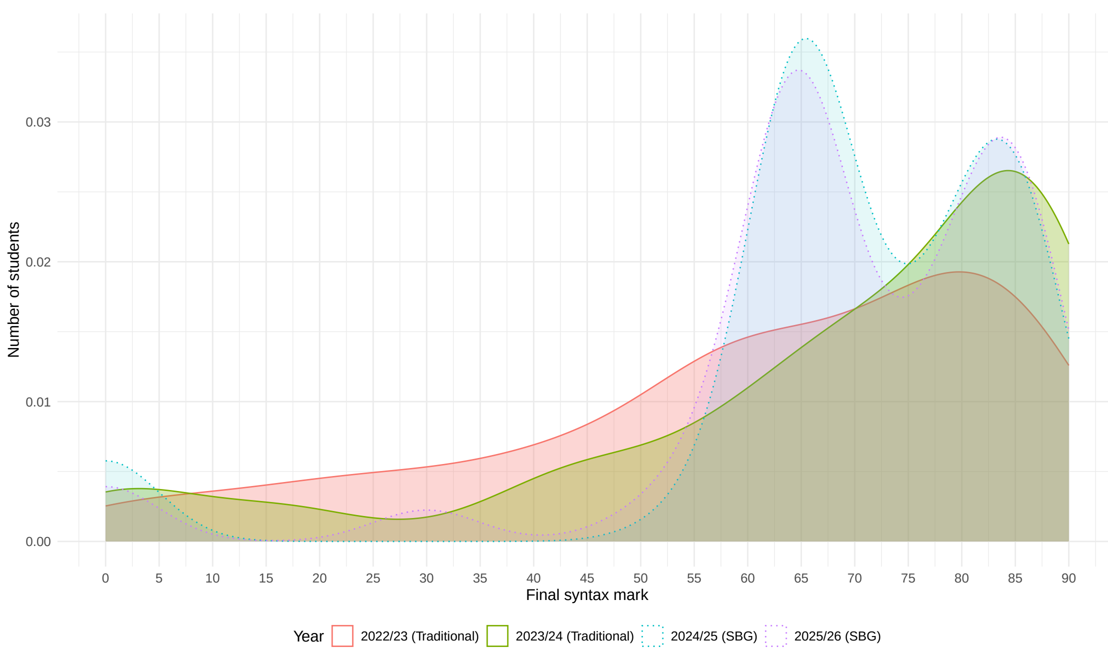


Figure 8: Number of students (y axis) who obtained the grades (x axis) in each of the past four iterations of the course.

C Attainment by gender

Marks in the syntax block, by year and gender: Figure 9. Marks in the syntax block compared to the other blocks of LEL2A ("baseline"), by gender: Figure 10.

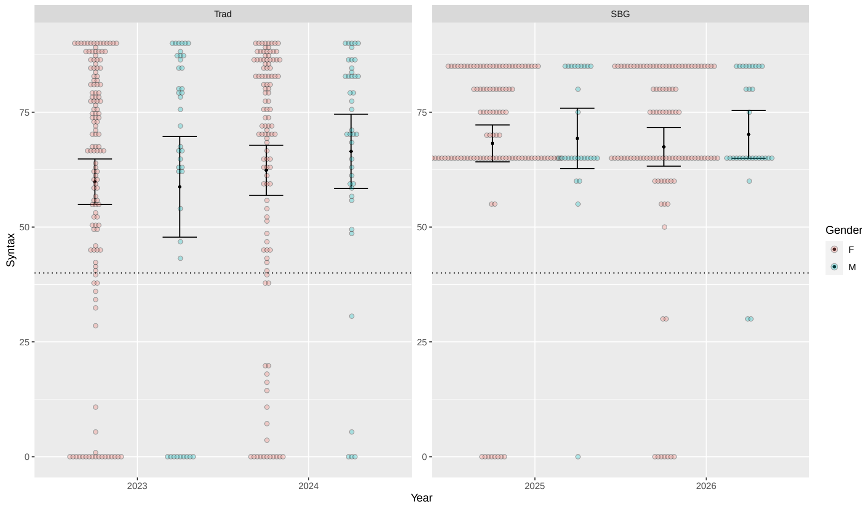


Figure 9: Mark distribution in the syntax block across years and gender. Dotted line is the passing mark.

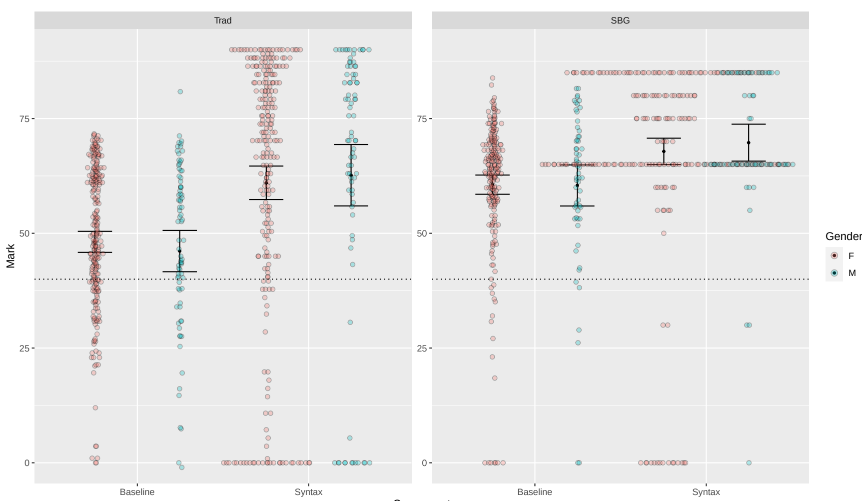


Figure 10: Mark distribution in different components of LEL2A (syntax vs baseline of other components) across years and gender. Dotted line is the passing mark.

D Assessment anxiety

We evaluated assessment anxiety using the TAI-5 survey Taylor and Deane (2002), replicating Lewis (2022) while adapting the post-test questions to the syntax block specifically.

Pre-test:

	1	2	3	4	5
The idea of doing assessments makes me feel very tense.	0	3	14	17	12
I wish assessments did not bother me so much.	1	6	7	15	17
I worry that I defeat myself while working on important assessments.	0	7	14	15	9
I feel very panicky when completing assessments.	1	5	14	18	8
During assessments I get so nervous that I forget things I really know.	2	11	14	9	10

Table 3: Pre-test TAI-5 anxiety survey results, from 1 (do not agree at all) to 5 (strongly agree). N = 46.

Post-test:

	1	2	3	4	5
Doing assessments on LEL2A syntax made me feel very tense.	3	11	3	0	1
I wish the LEL2A syntax assessment did not bother me so much.	8	5	2	2	1
I worry that I defeated myself while working on LEL2A syntax assessments.	6	10	0	1	1
I felt very panicky when completing LEL2A syntax assessments.	5	7	4	0	2
During LEL2A syntax assessments I got so nervous that I forgot things I really know.	11	4	1	0	2

Table 4: Post-test TAI-5 anxiety survey results, from 1 (do not agree at all) to 5 (strongly agree). N = 18.

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